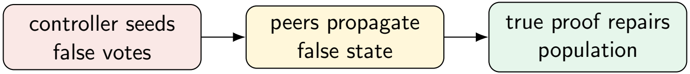
A mean-field guide for finding a persistent false-majority regime

Empirical peer propagation versus finite epistemic persistence

Study-design slides



The standalone question

Question

Can a controller create a false majority that keeps spreading through ordinary peer interaction after the controller is removed, while true proofs continue to circulate but are not perfectly persistent?

This is not the question "can the controller force everyone forever?"

We want the regime where the controller only creates the seed, and the population itself maintains or amplifies the false state.

All variables used in the model

Symbol	Meaning
x_t	fraction of agents voting for the false controller target at round t
ϕ_t	fraction of agents for whom the complete true proof is currently available at round t
q	number of visible peer slots in the prompt when no controller is present
L	number of true facts needed for a complete proof
ρ	probability that one true fact remains active from one round to the next
a	probability per round that an agent without a complete active proof acquires one
$g_q(x)$	measured probability that an incomplete-proof agent votes false after seeing q peer slots, when the population false share is x
A_q	strongest peer amplification of an already macroscopic false state

No other variables are needed for the main decision rule.

Runtime meaning of q

Important clarification

In the LLM game, q is simply the number of visible social peer slots.

$q = 2$ **controlled update**

$\boxed{\text{controller} + \text{peer}}$

$q = 3$ **controlled update**

$\boxed{\text{controller} + \text{peer} + \text{peer}}$

The LLM is not assumed to obey a classical unanimity rule. Therefore we do not assume a social term x^q in the final model.

Assumptions of the problem

1. There is one false semantic target Z and one true answer.
2. x_t is the population share voting for Z.
3. ϕ_t is the share of agents with all L true proof facts active.
4. Agents with complete active proof are strongly truth-directed.
5. Agents without complete proof can be influenced by peer votes.
6. Each true fact remains active with probability ρ per round.
7. New complete proofs are acquired with effective probability a per round.
8. The controller acts only to create an initial false seed. The post-release question sets the controller to NOOP.

The theorem is about post-release persistence: after seeding, can peers maintain the false majority?

Where the problem comes from

- Study 09b showed: the controller can create transient false majorities.
- But once complete proofs propagate, truth wins by the final round.
- So the actuator is not dead; the current game makes true information effectively absorbing.

Core diagnosis

With permanent information, the model eventually drives $\phi_t \to 1$. Then almost everyone has the complete proof, and the false state collapses.

We need finite epistemic persistence, not arbitrary lying, to make truth and social false propagation compete indefinitely.

Sanity check: permanent truth must win

If information is permanent, then

$$\rho = 1.$$

A complete proof, once active, remains active forever.

The mean-field update for complete-proof availability will be

$$\phi_{t+1} = \rho^L \phi_t + a(1 - \phi_t).$$

If $\rho = 1$, then $\rho^L = 1$, so

$$\phi_{t+1} = \phi_t + a(1 - \phi_t).$$

Therefore

$$1 - \phi_{t+1} = (1 - a)(1 - \phi_t).$$

Since $0 < a < 1$,

$$1 - \phi_t \to 0, \qquad \text{hence} \qquad \boxed{\phi_t \to 1.}$$

Sanity check: finite persistence leaves susceptible agents

Use the rough Study-09b calibration

$$L = 3, \qquad a \approx 0.24.$$

At stationarity,

$$\phi^* = \frac{a}{1 - \rho^L + a}.$$

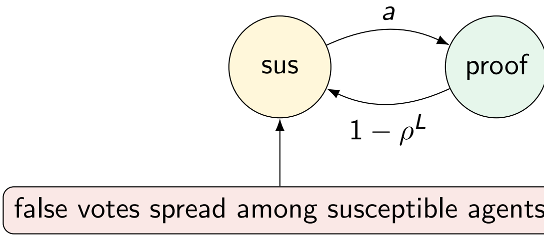
ρ	ρ^3	ϕ^*	susceptible share $1 - \phi^*$
1.00	1.000	1.000	0.000
0.95	0.857	0.628	0.372
0.90	0.729	0.470	0.530
0.85	0.614	0.384	0.616

Even high per-fact persistence can leave many agents without the complete proof, because all L facts must be active simultaneously.

A useful analogy: SIS with complex contagion

SIS intuition

- agents can become "protected" by complete proof;
- protected status can be lost if proof is not persistent;
- false votes can spread socially.



The analogy is not exact SIS: because peer influence can require a macroscopic seed, the false state may have a critical-mass threshold.

The bottleneck in the obvious approach

A tempting model is

$$\text{false social growth} \propto x^q.$$

This is the classical q-voter approximation.

Why this is the bottleneck

In the actual LLM runtime, q is only the number of visible peer slots. The LLM is not forced to switch only when all peers agree.

So a blind x^q theory can point to the wrong region. It imports a classical mechanism that may not match LLM social reasoning.

Main idea that fixes the bottleneck

$$\boxed{\text{Replace the assumed social law } x^q \text{ by the measured peer response } g_q(x).}$$

For each q, measure how an incomplete-proof agent reacts to peer contexts.
For example, for $q = 2$ we need only

$$p_{2,0}, \qquad p_{2,1}, \qquad p_{2,2},$$

where $p_{2,k}$ is the probability the focal agent votes false after seeing k false peers out of 2.

Then the theory uses the LLM's actual social response, not a classical rule.

High-level solution plan

1. Model complete-proof availability using one persistence parameter ρ.
2. Solve exactly for the long-run full-proof fraction ϕ^*.
3. Measure peer amplification $g_q(x)$ from LLM transitions.
4. Compress the relevant peer effect into one number A_q.
5. Prove that false-majority survival after release requires

$$\boxed{A_q(1-\phi^*) > 1.}$$

6. Convert this into a critical persistence value ρ_c.
7. Run experiments only near that predicted boundary.

Fact 1: exact solution for proof availability

Let

$$P = \rho^L$$

be the probability that the complete active proof survives one round.
The update is

$$\phi_{t+1} = P\phi_t + a(1 - \phi_t).$$

At stationarity, $\phi_{t+1} = \phi_t = \phi^*$, so

$$\begin{aligned}\phi^* &= P\phi^* + a(1 - \phi^*) \\ &= P\phi^* + a - a\phi^*.\end{aligned}$$

Move the ϕ^* terms to the left:

$$\phi^* - P\phi^* + a\phi^* = a.$$

Factor:

$$\phi^*(1 - P + a) = a.$$

Substitute $P = \rho^L$:

$$\boxed{\phi^* = \frac{a}{1 - \rho^L + a}.}$$

Fact 2: empirical peer response

For a focal agent without the complete proof, define

$$p_{q,k} = \Pr(\text{vote false next} \mid k \text{ false peers among } q).$$

If peers are sampled from a population with false share x, then the chance of seeing exactly k false peers is

$$\binom{q}{k} x^k (1-x)^{q-k}.$$

Therefore the measured peer-response curve is

$$\boxed{g_q(x) = \sum_{k=0}^{q} \binom{q}{k} x^k (1-x)^{q-k} p_{q,k}.}$$

For $q = 2$:

$$g_2(x) = (1-x)^2 p_{2,0} + 2x(1-x)p_{2,1} + x^2 p_{2,2}.$$

Fact 3: the post-release false-share map

After the controller is released, we ask whether peers alone maintain the false state. Only agents without the complete proof are susceptible. Their fraction is approximately

$$1 - \phi_t.$$

Among them, the probability of voting false after peer exposure is

$$g_q(x_t).$$

So the minimal post-release map is

$$\boxed{x_{t+1} \approx (1 - \phi_t) g_q(x_t).}$$

At long times, replace ϕ_t by ϕ^*:

$$\boxed{x_{t+1} \approx (1 - \phi^*) g_q(x_t).}$$

Fact 4: define majority-level amplification

We care about false majorities, so look only at

$$x \in [1/2, 1].$$

Define

$$\boxed{A_q = \max_{x \in [1/2,1]} \frac{g_q(x)}{x}.}$$

Interpretation

A_q is the largest multiplicative boost ordinary peers can give to an already macroscopic false state.

If $A_q \leq 1$, peers never amplify a false majority.
If $A_q > 1$, there is at least one majority-level false state where peer interaction increases false support among susceptible agents.

Detailed proof of the survival criterion

A false majority can sustain itself only if there exists some $x \geq 1/2$ such that

$$x_{t+1} > x_t.$$

Using the post-release map,

$$(1 - \phi^*) g_q(x) > x.$$

Divide both sides by $x > 0$:

$$(1 - \phi^*) \frac{g_q(x)}{x} > 1.$$

The largest possible value of $g_q(x)/x$ over false majorities is A_q.
Therefore such an x can exist only if

$$\boxed{A_q(1 - \phi^*) > 1.}$$

Conversely, if this inequality holds, the maximizing x gives local one-step growth.

Proof of the critical persistence formula I

Start from the survival criterion:

$$A_q(1-\phi^*) > 1.$$

Use

$$1-\phi^* = \frac{1-\rho^L}{1-\rho^L+a}.$$

At the boundary:

$$A_q \frac{1-\rho^L}{1-\rho^L+a} = 1.$$

Let

$$y = 1-\rho^L.$$

Then the boundary is simply

$$A_q \frac{y}{y+a} = 1.$$

This substitution is only to make the algebra one-line.

Proof of the critical persistence formula III

Starting from

$$A_q \frac{y}{y+a} = 1,$$

multiply both sides by $y + a$:

$$A_q y = y + a.$$

Move y to the left:

$$A_q y - y = a.$$

Factor y:

$$(A_q - 1)y = a.$$

Divide by $A_q - 1$:

$$\boxed{y = \frac{a}{A_q - 1}.}$$

Proof of the critical persistence formula II

From the previous slide,

$$y = \frac{a}{A_q - 1}.$$

But

$$y = 1 - \rho^L.$$

Therefore

$$1 - \rho^L = \frac{a}{A_q - 1}.$$

Move terms:

$$\rho^L = 1 - \frac{a}{A_q - 1}.$$

Take the L-th root:

$$\boxed{\rho_c = \left(1 - \frac{a}{A_q - 1}\right)^{1/L}.}$$

The formula is meaningful only when the quantity inside the root lies in $(0, 1)$, in particular $A_q > 1 + a$.

Endpoint cases and what they mean

Case 1: permanent information. If $\rho = 1$, then $\phi^* = 1$, so

$$A_q(1 - \phi^*) = 0.$$

A post-release false majority cannot persist.

Case 2: peer amplification too weak. The critical-persistence formula requires

$$A_q > 1 + a.$$

If $A_q \leq 1 + a$, persistence alone is not predicted to create the desired regime.

Case 3: hard upper bound. For $x \geq 1/2$, we always have $g_q(x) \leq 1$, so

$$A_q = \max_{x \in [1/2, 1]} \frac{g_q(x)}{x} \leq 2.$$

With $L = 3$ and $a \approx 0.242$, even this best case gives

$$\boxed{\rho_c \leq 0.912.}$$

Thus $\rho > 0.912$ is too persistent in this minimal closure.

Numerical examples: where to search

Use

$$L = 3, \qquad a \approx 0.242.$$

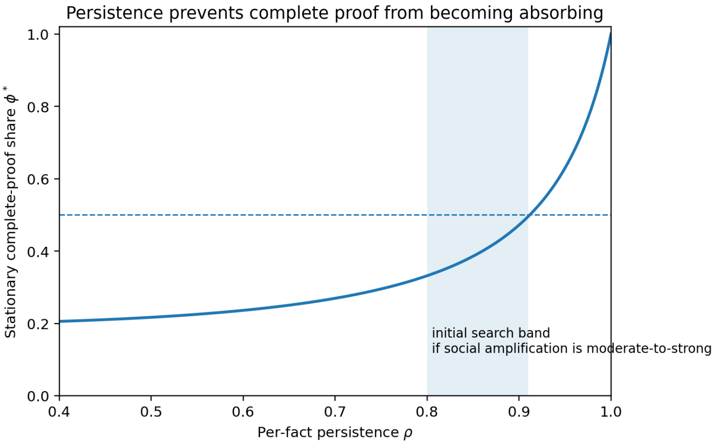
Then

$$\rho_c = \left(1 - \frac{0.242}{A_q - 1}\right)^{1/3}.$$

Measured A_q	Predicted ρ_c	Interpretation
1.30	0.578	weak peer amplification
1.50	0.802	moderate amplification
1.75	0.878	strong amplification
2.00	0.912	strongest possible majority-level amplification

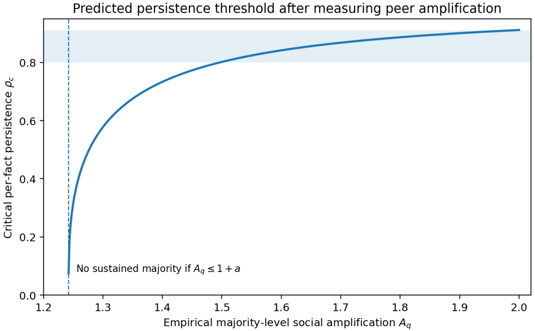
If A_2 is about 1.5–2, the first search region is roughly $\rho = 0.80$–0.91.

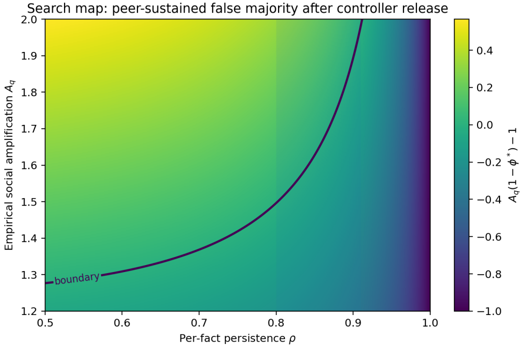
Picture 1: proof availability versus persistence



Lower persistence maintains a susceptible population; the useful regime is intermediate persistence, not complete forgetting.

Picture 2: predicted boundary





The expensive LLM search should happen only near the predicted boundary, after measuring A_2.

What to do next experimentally

1. Estimate a from proof-propagation trajectories.
2. Estimate A_2 from peer-only or no-controller transitions at $q = 2$.
3. Compute

$$\rho_c = \left(1 - \frac{a}{A_2 - 1}\right)^{1/L}.$$

4. Run a seed-and-release bracket:

$$\rho \in \{1, \rho_c + 0.03, \rho_c, \rho_c - 0.03\}.$$

5. Keep $q = 2$, $L = 3$, $N = 12$, and the controller settings fixed.
6. Only after a peer-sustained false regime is found, vary b/N to study efficiency.

Final main results

The revised theory has one central result. Let A_q measure how strongly ordinary peers amplify an already macroscopic false state, and let ρ measure how persistently each true fact remains available. Complete proof no longer becomes absorbing when $\rho < 1$; instead, the stationary fully informed fraction is $\phi^* = a/(1 - \rho^L + a)$. A controller-created false majority can sustain itself after the controller is removed only if peer amplification overcomes the remaining fully informed population: $A_q(1 - \phi^*) > 1$. This gives the closed-form threshold $\rho_c = (1 - a/(A_q - 1))^{1/L}$. Importantly, A_q is measured from the LLM peer transitions; it is not another parameter to sweep.

For the current $L = 3$ setup and the rough Study-09b proof-acquisition estimate $a \approx 0.24$, the first useful region is likely $\rho \approx 0.80$–0.91 if peer amplification is moderate to strong. Therefore the next decision should be: estimate A_2, calculate ρ_c, and run only a few seed-and-release simulations near that predicted boundary. If the false majority survives after controller release, then agents - not continued external control - are propagating the false collective state faster than epistemic repair can remove it.